



A tight lower bound on non-adaptive group testing estimation

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ARTICLE INFO

Article history:

Received 6 December 2023

Received in revised form 14 December 2024

Accepted 8 January 2025

Available online 17 January 2025

Keywords:

Group testing

Defect detection

Non-adaptive algorithms

Algorithm lower bounds

ABSTRACT

Efficiently counting or detecting defective items is a crucial task in various fields ranging from biological testing to quality control to streaming algorithms. The *group testing estimation problem* concerns estimating the number of defective elements d in a collection of n total within a given factor. We primarily consider the classical query model, in which a query reveals whether the selected group of elements contains a defective one. We show that any non-adaptive randomized algorithm that estimates the value of d within a constant factor requires $\Omega(\log n)$ queries. This confirms that a known $O(\log n)$ upper bound by Bshouty (2019) is tight and resolves a conjecture by Damaschke and Sheikh Muhammad (2010). Additionally, we prove similar matching upper and lower bounds in the threshold query model.

Published by Elsevier B.V.

1. Introduction

The *group testing problem* is a fundamental computational problem that concerns counting or detecting a set of defective items. Suppose there are d defective items in a collection of n total. Under the group testing model, a query can determine whether a selected group of elements contains a defective one. The concept was pioneered by Dorfman [20] to obtain more efficient syphilis testing by mixing blood samples together to perform tests in groups, and since then has exploded into a richly studied area. There are now various models considered [2,3,6,9,13,17], each with different applications as diverse as DNA testing [4,15,24], learning theory [21,23,26,27], and industrial processes [32]. Readers may see [2, Section 1.7] for a more extensive discussion.

In this work, we focus on *estimating* the number of defective items d in the randomized estimation setting. A randomized algorithm makes queries that may depend on randomness and outputs a valid answer with probability at least $2/3$. For a given factor⁴ $\alpha = 1 + \Omega_n(1)$, we say d^* is an α -estimation for d if $d \leq d^* \leq \alpha d$.

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¹ Supported by the Rényi Intézet Lendület Automorphic Research Group and NKFIH (National Research, Development and Innovation Office), Hungary grant K 143876.

² Research supported by an NSERC, Canada grant.

³ Supported by US National Science Foundation award 2006443.

⁴ The notation $\Omega_n(1)$ indicates that α may depend on n . Henceforth, we will omit the subscript for brevity.

An important distinction for query problems is the *adaptiveness* of algorithms. Adaptive algorithms may use the results of prior queries to choose the future ones. Nearly-tight bounds are known in the adaptive randomized estimation setting: Bshouty et al. [8] provided a randomized algorithm that returns a value d^* with $(1 - \varepsilon)d \leq d^* \leq (1 + \varepsilon)d$, and makes at most $\log \log d + \log^* n + O(1/\varepsilon^2)$ queries, where $\log^*$ is the iterated logarithm function. They also gave a nearly matching query complexity lower bound of $\log \log d + \Omega(1/\varepsilon)$.

Our work considers the *non-adaptive* case, where all queries are chosen at once. Despite being less powerful, non-adaptive algorithms have the benefits of parallelized testing and simpler test designs, which are significant for efficient implementation in real-world applications. Bshouty provided a polynomial-time $O(\log n)$ query constant-estimation algorithm [6] using ideas similar to [18,22], and it was conjectured to be tight [17]. A close lower bound of $\Omega(\log n / \log \log n)$ was independently proven in [6,31]. Very recently, Bshouty posted a manuscript that established the lower bound $\Omega(\log n / \log \log \dots \log n)$ for any constant k [7]; this is the same bound errantly claimed earlier [5]. In this work, we confirm that the conjectured $\Omega(\log n)$ bound is indeed tight.

Theorem 1.1. *Let $\alpha = 1 + \Omega(1)$. Any non-adaptive randomized algorithm that α -estimates the number of defective items in a set of n total must perform $\Omega(\log_\alpha n)$ queries.*

Note Theorem 1.1 holds for any estimation factor $\alpha = 1 + \Omega(1)$, even in super-constant regimes. For example, it states that a $\log n$ -estimation algorithm would require $\Omega(\log n / \log \log n)$ queries, illustrating that α -estimation is roughly as difficult as constant estimation unless α is quasi-polynomial in n . It should be noted, however, that this lower bound already follows from the techniques of [6,7] when $\alpha = \Omega(\log n)$ and $\alpha = \Omega(\log \log \dots \log n)$ for any constant k , respectively.

The main result follows from Theorem 1.2 as a special case under the *threshold query* model in the *promise* setting. Introduced in [16] and well-studied since then [1,11,12,25], the threshold query model is defined in the absolute sense: for a fixed threshold parameter $\lambda \in [n]$, the query oracle returns 1 on a set of items Q if and only if Q contains at least λ defective items. Unlike some prior works, we do not distinguish between constant and non-constant thresholds in our lower bound. In the threshold model framework, the classical query oracle is simply a 1-threshold.

Notice that the threshold query model cannot produce a good prediction when the number of defective items is below the threshold. Specifically for $\lambda \geq 3$, the λ -threshold query result on a collection with 1 or $\lambda - 1$ defective items is identically 0 regardless of the query size. To exclude the vacuous cases, it is reasonable to assume that the collection contains at least λ defective items. More generally, for non-negative integers L and U with $L < U$, the $[L, U]$ -*promise* states that the number of defective items d satisfies $L \leq d \leq U$.

We show a logarithmic query complexity lower bound in terms of the promise gap for the promise estimation problem with threshold queries. The lower bound for the standard query model follows by setting $\lambda = 1$, $L = 1$, and $U = n$, which attains the claimed bound of $\Omega(\log_\alpha n)$.

Theorem 1.2. *Let $\alpha = 1 + \Omega(1)$ and $\lambda \in [n]$. Suppose L and U are integers that satisfy $\lambda \leq L < U \leq n$. Under the $[L, U]$ -promise, any non-adaptive randomized α -estimation algorithm with λ -threshold queries must perform $\Omega(\log_\alpha(U/L))$ queries.*

In Section 4 we prove this result is tight for any α and constant λ by extending the upper bound in [6] to the general threshold case.

1.1. Distribution distinguishing problem

A crucial idea of both prior work [6,7] and our proof is the connection with the *distribution distinguishing* problem, an algorithmic formulation of hypothesis testing. The general setting of the problem is as follows: given two probability distributions μ_0 and μ_1 , a uniform bit $b \in \{0, 1\}$ is chosen, then a sample x is drawn from the distribution μ_b . The objective is to determine which distribution the sample x was drawn from. Intuitively, the distribution distinguishing problem is hard if the two distributions are close, in the sense that $\mu_0(A) \approx \mu_1(A)$ for every subset A in the probability space.

The earlier works of [6,7] make use of the distribution distinguishing problem by showing that if too few queries are made, a vacant set of queries would be able to distinguish between two distributions (i.e. solve the distribution distinguishing problem). More precisely, for a non-adaptive, α -estimation algorithm $\mathcal{A}$, they partition the random q queries of $\mathcal{A}$ into b buckets depending on the query sizes, and show that there exists a specific (i.e., the choice does not depend on the randomness) bucket $\mathcal{B}$ that contains $O(q/b)$ queries with high probability. Moreover, $\mathcal{B}$ determines two distributions on sets of defective items that $\mathcal{A}$ can distinguish between, but where (essentially) the only informative queries are in $\mathcal{B}$. In other words, there is some collection of query sizes and two distributions, such that the majority of queries useful for distinguishing between those distributions must be one of the sizes in the collection, and yet it is likely the algorithm $\mathcal{A}$ has few queries of this size. If $q = o(b)$, then $\mathcal{B}$ will typically contain zero queries, so $\mathcal{A}$ cannot distinguish between distributions, leading to a contradiction. The earlier paper [6] carries out this strategy with $b = \Theta(\log n / \log \log n)$, whereas the recent work [7] performs a more sophisticated iteration process to obtain the strengthened bound $\Omega(\log n / \log \log \dots \log n)$ for any constant k .

Our proof circumvents the need to design bucket-dependent distributions and further complications, and simply adopts a pair of distributions “naturally distinguishable” by an α -estimation algorithm. We start by constructing a pair of distributions $(\mu_{\text{even}}, \mu_{\text{odd}})$ on the planted set of defective items with two properties. The first is that the distributions should have sufficiently disjoint supports. That is, no element sampled from μ_{even} can have a size within an α -factor of the size of an element sampled from μ_{odd} . Thus, any algorithm which can α -estimate the number of defective items must be able to distinguish between the two distributions. The second is that without making a large number of queries, the distributions *induced by the queries* should be difficult to distinguish. Intuitively, the hardness requirement is related to how close the induced distributions are, and this is formally quantified by the total variation distance.

More precisely, we choose the distributions to “multiplicatively interleave” the possible support sizes, where μ_{even} is a distribution over sets of size even powers of $\lfloor \alpha \rfloor + 1$ and μ_{odd} over odd powers. Observe we are essentially viewing the number of defective items on a logarithmic scale. In order to correctly approximate this number, any non-adaptive algorithm must choose the appropriate scale; otherwise, the query results will be identically 0 or 1 with high probability. We randomly plant defective items such that if $o(\log n)$ queries are made, it is unlikely the correct scale is chosen, and thus little information can be learned.

Our analysis relies on the powerful *coupling* technique, which we believe may have more applications for related problems. We will formalize the notions of distribution distance and query-induced distributions in Section 2.

Concurrent work. This paper is a combination of independent and concurrent work by Bshouty and Harcos [10] and Cheung, Hatami, and Ostuni [14], both of which proved [Theorem 1.1](#) using similar techniques.

Overview. In Section 2, we lay out several tools and technical estimates that are required to prove our results, including the notion of query-induced distributions in Section 2.1, a refresher on the hypergeometric distribution in Section 2.2, and fundamental facts about total variation distance and coupling in Section 2.3. Section 3 contains the proof of [Theorem 1.2](#). In Section 4, we present a query algorithm that achieves the tight upper bound in the threshold query model.

2. Preliminaries

2.1. Notations

We denote $[n]$ for $\{1, \dots, n\}$, and adopt the standard computer science asymptotic notations.

We introduce the following notations for the threshold query model. In the non-adaptive setting, q queries are specified by a collection of subsets $Q_1, \dots, Q_q \subseteq [n]$, such that Q_i does not depend on the responses to other queries. For a set of defective items $B \subseteq [n]$ and the query set $W \subseteq [n]$, we use the notation $W^{\geq \lambda}(\cdot)$ to denote the query result on W :

$$W^{\geq \lambda}(B) = \begin{cases} 1 & \text{if } |W \cap B| \geq \lambda \\ 0 & \text{otherwise.} \end{cases}$$

For $\bar{Q} = (Q_1, \dots, Q_q)$, we use the shorthand $\bar{Q}^{\geq \lambda}(B) = (Q_1^{\geq \lambda}(B), \dots, Q_q^{\geq \lambda}(B)) \in \{0, 1\}^q$ to denote the tuple of q queries. For a distribution μ on subsets of $[n]$, the *induced distribution* is the distribution of $\bar{Q}^{\geq \lambda}(B)$ for B sampled from μ .

2.2. Hypergeometric distribution

The *hypergeometric distribution* $\mathcal{H}_{n,k,s}$ is naturally associated with the group testing problem. It is characterized by three parameters:

- n : the total number of items;
- k : the number of marked items;
- s : the number of items sampled in a uniform draw (without replacement).

The number of marked items sampled in the draw is given by the distribution:

$$\Pr[r \text{ marked items sampled}] = \frac{\binom{k}{r} \binom{n-k}{s-r}}{\binom{n}{s}}.$$

We adopt the convention that $\binom{N}{R} = 0$ whenever $R > N$ or $R < 0$.

It is a well-known fact that the hypergeometric random variable $H_{n,k,s} \sim \mathcal{H}_{n,k,s}$ has mean ks/n . For the proof of the main result, we will need the following concentration inequalities for hypergeometric distributions. Markov's inequality implies that

$$\Pr[H_{n,k,s} \geq \gamma] \leq \frac{ks}{\gamma n}. \quad (1)$$

A Chernoff-type lower tail bound for hypergeometric distributions is implicit in [28]: for $\xi < ks/n$,

$$\Pr[H_{n,k,s} \leq \xi] \leq \exp\left(-\frac{(ks/n - \xi)^2}{2ks/n}\right). \quad (2)$$

2.3. Total variation distance and coupling

As mentioned in Section 1.1, the key technique of our proof is to construct two close distributions of sets of defective items. The closeness is quantified by the *total variation distance*.

Definition 2.1 (*Total variation distance*). Let μ_0 and μ_1 be two probability measures on the measurable space $(S, \mathcal{F})$. The total variation distance of the two measures is

$$\|\mu_0 - \mu_1\|_{\text{TV}} := \sup_{A \in \mathcal{F}} |\mu_0(A) - \mu_1(A)|.$$

The above notion provides a mathematical interpretation of statistical closeness. From the algorithmic perspective, statistical closeness can be captured by the hardness of the distribution distinguishing problem. More precisely, the total variation distance can be viewed as a measure of how well the optimal algorithm for the distinguishing problem (also called *distinguisher*) outperforms a random guess.

Let μ_0 and μ_1 be two probability measures on the measurable space $(S, \mathcal{F})$. We say that $\Psi : S \rightarrow \{0, 1\}$ is a deterministic distinguisher between μ_0 and μ_1 if

$$\Pr_{\substack{b \sim \{0,1\} \\ x \sim \mu_b}} [\Psi(x) = b] \geq \frac{2}{3}$$

where $b \sim \{0, 1\}$ means that b is chosen uniformly at random from $\{0, 1\}$.

Lemma 2.2. Let μ_0 and μ_1 be two probability measures on the measurable space $(S, \mathcal{F})$. Let $\Psi : S \rightarrow \{0, 1\}$ be any deterministic distinguisher between μ_0 and μ_1 . Then

$$\frac{1}{3} \leq 2 \left(\Pr_{\substack{b \sim \{0,1\} \\ x \sim \mu_b}} [\Psi(x) = b] - \frac{1}{2} \right) \leq \|\mu_0 - \mu_1\|_{\text{TV}}.$$

Proof. Let $A = \{x \in S : \Psi(x) \neq 0\}$ be the support of Ψ . Then

$$\frac{1}{6} \leq \Pr_{\substack{b \sim \{0,1\} \\ x \sim \mu_b}} [\Psi(x) = b] - \frac{1}{2} = \frac{1}{2} \left(\Pr_{x \sim \mu_0} [x \in A] + \Pr_{x \sim \mu_1} [x \notin A] - 1 \right) = \frac{1}{2} (\mu_0(A) - \mu_1(A)) \leq \frac{1}{2} \|\mu_0 - \mu_1\|_{\text{TV}}. \quad \square$$

By averaging, the above bound also holds for randomized distinguishers.

Often exactly computing the total variation distance is difficult. A fundamental connection with the notion of *coupling* provides a way to upper bound the total variation distance.

Definition 2.3 (*Coupling*). Let μ_1 and μ_2 be two probability measures on the measurable space $(S, \mathcal{F})$. A coupling of μ_1 and μ_2 is a probability measure ζ on the product space $(S \times S, \mathcal{F} \times \mathcal{F})$ such that

$$\zeta(A \times S) = \mu_1(A) \text{ and } \zeta(S \times A) = \mu_2(A) \text{ for all } A \in \mathcal{F}.$$

The following is a standard inequality (see e.g. [30, Section 4.1.3]). We include a proof for completeness.

Theorem 2.4 (*Coupling inequality*). Let μ_1 and μ_2 be two probability measures on the measurable space $(S, \mathcal{F})$. For any coupling ζ of μ_1 and μ_2 ,

$$\|\mu_1 - \mu_2\|_{\text{TV}} \leq \Pr_{(X,Y) \sim \zeta} [X \neq Y].$$

Proof. For any $A \in \mathcal{F}$, we have

$$\begin{aligned} \mu_1(A) - \mu_2(A) &= \Pr[X \in A] - \Pr[Y \in A] \\ &= \Pr[X \in A, X = Y] + \Pr[X \in A, X \neq Y] - \Pr[Y \in A, X = Y] - \Pr[Y \in A, X \neq Y] \\ &= \Pr[X \in A, X \neq Y] - \Pr[Y \in A, X \neq Y] \\ &\leq \Pr[X \neq Y]. \quad \square \end{aligned}$$

Readers unfamiliar with the concept of coupling may find it helpful to consult [Appendix A](#) for additional background. (Those comfortable with standard coupling manipulations may safely skip that section.)

3. Proof of Theorem 1.2

This section is dedicated to the proof of Theorem 1.2. To recap the setting, for a fixed $\lambda \in [n]$, the λ -threshold query oracle detects whether a query set contains at least λ defective items. The promise that the collection contains at least L ($\geq \lambda$) and at most U defective items is given.

We construct a pair of distributions that is hard for any α -estimation algorithm to distinguish. Let $\beta = \lfloor \alpha \rfloor + 1$, and define

$$\begin{aligned} S_{\text{even}} &= \{L\beta^2, L\beta^4, \dots, L\beta^{2m}\}, \\ S_{\text{odd}} &= \{L\beta, L\beta^3, \dots, L\beta^{2m-1}\}, \end{aligned}$$

where m is the largest integer which $L\beta^{2m} \leq U$. It is clear that $m = \Theta(\log_\beta(U/L)) = \Theta(\log_\alpha(U/L))$.

The distribution μ_{even} (resp. μ_{odd}) is constructed by the following sampling procedures:

1. Sample s from S_{even} (resp. S_{odd}) uniformly at random;
2. Sample a set of s items uniformly at random.

We claim that if the deterministic queries $\vec{Q}^{\geq \lambda}$ output a valid estimation, the same queries distinguish $\vec{Q}^{\geq \lambda}(\mu_{\text{odd}})$ and $\vec{Q}^{\geq \lambda}(\mu_{\text{even}})$ without error. Indeed, since $\alpha < \beta$ we have $\alpha L\beta^{2i} < L\beta^{2i+1} < \alpha^{-1}L\beta^{2i+2}$ and therefore $[L\beta^{2i}, \alpha L\beta^{2i+1}] \cap [L\beta^{2j}, \alpha L\beta^{2j+1}] = \emptyset$ for all distinct $i, j \in [m]$. Thus, there is exactly one $s \in S_{\text{odd}} \cup S_{\text{even}}$ for which the estimation algorithm output is within the range of $[s, \alpha s]$, so this algorithm is a deterministic distinguisher between μ_{even} and μ_{odd} . Hence Lemma 2.2 implies

$$\|\vec{Q}^{\geq \lambda}(\mu_{\text{even}}) - \vec{Q}^{\geq \lambda}(\mu_{\text{odd}})\|_{\text{TV}} \geq \frac{1}{3}. \quad (3)$$

It remains to derive an upper bound for $\vec{Q}^{\geq \lambda}(\mu_{\text{even}})$ and $\vec{Q}^{\geq \lambda}(\mu_{\text{odd}})$ in terms of the number of queries q . In the next lemma, we show that for any deterministic queries, the total variation distance of the induced distributions is proportional to $\frac{1}{\log_\alpha(U/L)}$ and the number of queries.

Lemma 3.1. Let $\vec{Q}^{\geq \lambda} = (Q_1^{\geq \lambda}, \dots, Q_q^{\geq \lambda})$ be q deterministic queries with the threshold query oracle. Then

$$\|\vec{Q}^{\geq \lambda}(\mu_{\text{even}}) - \vec{Q}^{\geq \lambda}(\mu_{\text{odd}})\|_{\text{TV}} = O\left(\frac{q}{\log_\alpha(U/L)}\right).$$

Moreover, a randomized query algorithm can be viewed as a distribution over deterministic queries. We formally establish this fact and include the proof for completeness.

Lemma 3.2. If there is a randomized non-adaptive algorithm that asks q queries and for every set of defective items I , with probability at least $2/3$, outputs an α -estimation of $|I|$, then there are q queries $\vec{Q}^{\geq \lambda} = (Q_1, \dots, Q_q)$ and a deterministic distinguisher Ψ between $\vec{Q}^{\geq \lambda}(\mu_{\text{even}})$ and $\vec{Q}^{\geq \lambda}(\mu_{\text{odd}})$.

Proof. Let $\mathcal{A}(s, I)$ be a randomized non-adaptive algorithm that asks q queries and for any set of defective items I , with probability at least $2/3$, outputs an α -estimation of $|I|$, where s is the random seed of the algorithm.

Consider the indicator random variable $X(s, I)$, which equals 1 if the estimation is correct. For every I , we have $\mathbb{E}_s[X(s, I)] \geq 2/3$. Consider the distribution D of I where, with probability $1/2$, I is chosen according to μ_{odd} , and with probability $1/2$, it is chosen according to μ_{even} . Then

$$\mathbb{E}_s[\mathbb{E}_{I \sim D}[X(s, I)]] = \mathbb{E}_{I \sim D}[\mathbb{E}_s[X(s, I)]] \geq 2/3$$

and therefore there exists an s_0 such that $\mathbb{E}_{I \sim D}[X(s_0, I)] \geq 2/3$. In other words, there exists a set of q queries $\vec{Q}^{\geq \lambda} = (Q_1, \dots, Q_q)$ such that, for random I chosen according to D , with probability at least $2/3$, the deterministic algorithm $\mathcal{A}(s_0, I)$ correctly estimates $|I|$. The same algorithm can distinguish between $\vec{Q}^{\geq \lambda}(\mu_{\text{even}})$ and $\vec{Q}^{\geq \lambda}(\mu_{\text{odd}})$ because, since $\beta = \lfloor \alpha \rfloor + 1$, the output of the estimation algorithm, which falls within the range $[|I|, \alpha|I|]$, uniquely determines whether $|I| \in S_{\text{odd}}$ or $|I| \in S_{\text{even}}$. $\square$

Lemmas 3.1 and 3.2 combined with Eq. (3) imply that any randomized α -estimation algorithm using q queries must satisfy

$$\frac{1}{3} \leq \|\vec{Q}^{\geq \lambda}(\mu_{\text{even}}) - \vec{Q}^{\geq \lambda}(\mu_{\text{odd}})\|_{\text{TV}} \leq O\left(\frac{q}{\log_\alpha(U/L)}\right) \implies q = \Omega(\log_\alpha(U/L)),$$

and this completes the proof.

Proof of Lemma 3.1. We consider the coupling (X, Y) of μ_{even} and μ_{odd} defined by the following sampling procedures:

1. Sample $j \in [m]$ uniformly at random;
2. Sample a size- $L\beta^{2j}$ set uniformly at random to be X ;
3. Sample a size- $L\beta^{2j-1}$ set uniformly at random to be Y .

It is evident that (X, Y) is indeed a coupling for μ_{even} and μ_{odd} ; therefore $(\bar{Q}^{\geq \lambda}(X), \bar{Q}^{\geq \lambda}(Y))$ is a coupling for $\bar{Q}^{\geq \lambda}(\mu_{\text{even}})$ and $\bar{Q}^{\geq \lambda}(\mu_{\text{odd}})$ (see Lemma A.2). By the coupling inequality (Theorem 2.4) and a union bound, we have

$$\|\bar{Q}^{\geq \lambda}(\mu_{\text{even}}) - \bar{Q}^{\geq \lambda}(\mu_{\text{odd}})\|_{\text{TV}} \leq \Pr[\bar{Q}^{\geq \lambda}(X) \neq \bar{Q}^{\geq \lambda}(Y)] \leq \sum_{i=1}^q \Pr[Q_i^{\geq \lambda}(X) \neq Q_i^{\geq \lambda}(Y)].$$

It suffices to show that

$$\Pr[W^{\geq \lambda}(X) \neq W^{\geq \lambda}(Y)] = O\left(\frac{1}{\log_{\alpha}(U/L)}\right) \quad (4)$$

for any $W \subseteq [n]$. The intuition behind the coupling is that the sizes of X and Y are within a β -factor; consequently one may expect that the threshold query results are most likely equal for defect sets of comparable sizes. Towards proving Eq. (4), we actually prove the stronger statement that with high probability, $W^{\geq \lambda}(X) = W^{\geq \lambda}(Y) = 0$ when j is small, and $W^{\geq \lambda}(X) = W^{\geq \lambda}(Y) = 1$ when j is large.

The disagreement probability is an average of conditional probabilities in terms of j :

$$\Pr[W^{\geq \lambda}(X) \neq W^{\geq \lambda}(Y)] = \frac{1}{m} \sum_{j=1}^m \Pr[\{W^{\geq \lambda}(X), W^{\geq \lambda}(Y)\} = \{0, 1\} \mid |X| = L\beta^{2j}, |Y| = L\beta^{2j-1}].$$

Denote the conditional probability by P_j , and let $k = |W|$. Then $P_j = P_j^{(1)} + P_j^{(2)}$, where

$$P_j^{(1)} := \Pr[H_{n,k,L\beta^{2j}} \geq \lambda] \Pr[H_{n,k,L\beta^{2j-1}} < \lambda]$$

and

$$P_j^{(2)} := \Pr[H_{n,k,L\beta^{2j}} < \lambda] \Pr[H_{n,k,L\beta^{2j-1}} \geq \lambda].$$

Let $m_* = m_*(k) := \max\{j \in \mathbb{Z} : kL\beta^{2j}/n \leq \lambda\}$. We split the sum into the following three ranges:

$$J_{\text{Low}} = (-\infty, m_*] \cap [m], \quad J_{\text{Mid}} = \{m_* + 1\} \cap [m], \quad J_{\text{High}} = [m_* + 2, \infty) \cap [m].$$

We want to show that the overall sum of $P_j^{(1)}$ is $O(1)$ by showing that

$$\sum_{j \in J_{\text{Low}}} P_j^{(1)}, \sum_{j \in J_{\text{Mid}}} P_j^{(1)}, \sum_{j \in J_{\text{High}}} P_j^{(1)} = O(1), \quad (5)$$

and the sum of $P_j^{(2)}$ can be handled by a similar argument. We remark that some of these index sets may be empty, in this case, the corresponding sum is vacuously zero. For $j \in J_{\text{Mid}}$, the trivial bound $P_{m_*+1}^{(1)} \leq 1$ implies that $\sum_{j \in J_{\text{Mid}}} P_j^{(1)} \leq 1$ as desired.

For $j \in J_{\text{Low}}$, we use Markov's inequality (Eq. (1)):

$$P_j^{(1)} = \Pr[H_{n,k,L\beta^{2j}} \geq \lambda] \Pr[H_{n,k,L\beta^{2j-1}} < \lambda] \leq \Pr[H_{n,k,L\beta^{2j}} \geq \lambda] \leq \frac{kL\beta^{2j}}{\lambda n}.$$

From this, we can show that the contribution for $j \in J_{\text{Low}}$ is at most a constant:

$$\sum_{j \in J_{\text{Low}}} P_j^{(1)} \leq \frac{kL}{\lambda n} \sum_{j=1}^{m_*} \beta^{2j} = \frac{kL}{\lambda n} \cdot \frac{\beta^2(\beta^{2m_*} - 1)}{\beta^2 - 1} \leq O(1) \cdot \frac{kL\beta^{2m_*}}{\lambda n} = O(1).$$

For $j \in J_{\text{High}}$, we use the Chernoff-type bound (Eq. (2)):

$$P_j^{(1)} = \Pr[H_{n,k,L\beta^{2j}} \geq \lambda] \Pr[H_{n,k,L\beta^{2j-1}} < \lambda] \leq \Pr[H_{n,k,L\beta^{2j-1}} \leq \lambda] \leq \exp\left(-\frac{(kL\beta^{2j-1}/n - \lambda)^2}{2kL\beta^{2j-1}/n}\right).$$

By the definition of m_* and the fact that $\beta \geq 2$, we have, for every $j \in J_{\text{High}}$,

$$\frac{kL}{n} \beta^{2j-1} \geq \frac{kL}{n} \beta^{2(m_*+2)-1} = \beta \left(\frac{kL\beta^{2(m_*+1)}}{n} \right) > \beta\lambda \geq 2\lambda. \quad (6)$$

Therefore

$$P_j^{(1)} \leq \exp\left(-\frac{(kL\beta^{2j-1}/n - \lambda)^2}{2kL\beta^{2j-1}/n}\right) \leq \exp\left(-\frac{(kL\beta^{2j-1}/n - kL\beta^{2j-1}/(2n))^2}{2kL\beta^{2j-1}/n}\right) = \exp\left(-\frac{1}{8} \cdot \frac{kL\beta^{2j-1}}{n}\right).$$

Applying this bound and (6), we can show that the contribution for $j \in J_{\text{High}}$ is also at most a constant:

$$\begin{aligned} \sum_{j \in J_{\text{High}}} P_j^{(1)} &\leq \sum_{j=m_*+2}^m \exp\left(-\frac{1}{8} \cdot \frac{kL\beta^{2j-1}}{n}\right) \\ &= \sum_{j=m_*+2}^m \exp\left(-\frac{1}{8} \cdot \frac{kL\beta^{2(m_*+2)-1}}{n} \beta^{2(j-m_*-2)}\right) \\ &\leq \sum_{r=0}^{\infty} \exp\left(-\frac{\lambda\beta^{2r}}{4}\right) \\ &\leq \sum_{r=1}^{\infty} (e^{-\lambda/4})^r = O(1). \end{aligned}$$

This completes the justification of Eq. (5). Combining with an analogous argument for $P_j^{(2)}$, we have

$$\Pr[W^{\geq \lambda}(X) \neq W^{\geq \lambda}(Y)] = \frac{1}{m} \sum_{j=1}^m (P_j^{(1)} + P_j^{(2)}) = O\left(\frac{1}{m}\right) = O\left(\frac{1}{\log_{\alpha}(U/L)}\right). \quad \square$$

4. The upper bound

In [6], Bshouty presented a polynomial-time $O(\log n)$ query constant-estimation algorithm using ideas similar to [18, 22]. In this section, we expand upon this result to cover any α -estimation and the threshold query model for any α and constant λ . Specifically, we prove:

Theorem 4.1. *Let $\lambda \in \mathbb{N}$ be a constant and $\alpha = 1 + \Omega(1)$. Suppose L and U are integers that satisfy $\lambda \leq L < U \leq n$. Under the $[L, U]$ -promise, there is a non-adaptive randomized α -estimation algorithm that makes $O(\log_{\alpha}(U/L))$ λ -threshold queries.*

4.1. Definitions and preliminary results

In this section, we give some definitions and results we will need to prove Theorem 4.1.

A λ -threshold p -query is a query Q that contains each $i \in [n]$ randomly and independently with probability p . For any constant λ and given a λ -threshold p -query $Q^{\geq \lambda}$, we define

$$P_{\lambda}(d, p) := \Pr[Q^{\geq \lambda}(I) = 1] = 1 - \sum_{i=0}^{\lambda-1} \binom{d}{i} p^i (1-p)^{d-i} = \sum_{i=\lambda}^d \binom{d}{i} p^i (1-p)^{d-i}, \quad (7)$$

where $|I| = d$. For $\lambda = 1$, we define

$$P(d, p) := P_1(d, p) = \Pr[Q(I) = 1] = 1 - (1-p)^d,$$

where Q is a p -query (1-threshold p -query) and $|I| = d$.

We will use the following standard concentration inequality:

Lemma 4.2 (Chernoff's bound). *Let $X_1, \dots, X_t$ be independent random variables taking values $\{0, 1\}$. Let $X = X_1 + \dots + X_t$ and $\mathbb{E}[X] = \mu$. Then, for every $\eta \geq 0$,*

$$\Pr[X \geq (1 + \eta)\mu] \leq \left(\frac{e^{\eta}}{(1 + \eta)^{1+\eta}}\right)^{\mu} \leq e^{-\eta^2 \mu / (2 + \eta)}$$

and for every $0 < \eta < 1$,

$$\Pr[X \leq (1 - \eta)\mu] \leq \left(\frac{e^{-\eta}}{(1 - \eta)^{1-\eta}}\right)^{\mu} \leq e^{-\eta^2 \mu / 2}.$$

The following lemma enables us to assume that $d > d'$ for any constant d' independent of n .

Lemma 4.3. *Let $d' = O_n(1)$ be a constant. Any algorithm that α -estimates d assuming $d > d'$ with $O(\log(1/\delta) \log n)$ queries can be modified to an algorithm that α -estimates d for any d with $O(\log(1/\delta) \log n)$ queries.*

Proof. Let A be an algorithm that, with probability at least $1 - \delta/3$, α -estimates d assuming $d > d'$. We first want to augment A with $O(\log(1/\delta))$ queries of a non-adaptive algorithm that, with probability at least $1 - \delta/3$, accepts if $d \leq d'$ and rejects if $d \geq 8d'$. To achieve this, consider the λ -threshold p -query Q when $p = \lambda/(2d')$. If $d \leq d'$ then the expectation

of $|I \cap Q|$ is $dp \leq \lambda/2$. Therefore, by Markov's bound, the probability that the answer is 0 is at least $1/2$. If $d \geq 8d'$ then the expectation of $|I \cap Q|$ is $dp \geq 4\lambda$. By [Lemma 4.2](#), the probability that the answer to the query is 0 is

$$\Pr[|I \cap Q| < \lambda] \leq e^{-(3/4)^2 \cdot 4\lambda/2} \leq \frac{1}{3}.$$

Therefore, with $O(\log(1/\delta))$ λ -threshold p -queries, with probability at least $1 - \delta$, we can distinguish between $d \leq d'$ and $d \geq 4d'$.

Additionally, we incorporate into A the queries from the non-adaptive algorithm of Cheraghchi in [\[13\]](#) that, with probability at least $1 - \delta/3$, identifies all the defective items, assuming their count is less than $4d'$. By [\[13\]](#), the number of queries in the latter algorithm is $O(d^2 \log d' \log n) = O(\log n)$.

If the second algorithm accepts ($d \leq 4d'$), we employ the algorithm from [\[13\]](#) to identify all defective items. In particular, we find d exactly. If it rejects ($d > d'$) we run A to α -estimate d . $\square$

For the proof of [Theorem 4.1](#), we will need the following form of Chernoff's bound:

Lemma 4.4. Let $X_1, X_2, \dots, X_t$ be independent random variables that take values in $\{0, 1\}$. Let $X = (X_1 + X_2 + \dots + X_t)/t$ and $\mu \leq \mathbb{E}[X]$. Then for any $\Gamma \geq \mu$ we have

$$\Pr[X \geq \Gamma] \leq \left(\frac{e^{1-\frac{\mu}{\Gamma}} \mu}{\Gamma} \right)^{\Gamma t} \leq \left(\frac{e\mu}{\Gamma} \right)^{\Gamma t}.$$

Proof. Let $\mu' = \mathbb{E}[\sum X_i] = t\mathbb{E}[X]$. Applying Chernoff's bound ([Lemma 4.2](#)) with $\eta = \Gamma t/\mu' - 1$ yields

$$\Pr[X \geq \Gamma] \leq \left(\frac{e^{\left(\frac{\Gamma t}{\mu'} - 1\right)}}{\left(\frac{\Gamma t}{\mu'}\right)^{\Gamma t/\mu'}} \right)^{\mu'} = \left(\frac{e^{(1-\frac{\mu'}{\Gamma})} \mu'}{\Gamma t} \right)^{\Gamma t} = \left(\frac{e^{(1-\frac{\mathbb{E}[X]}{T})} \mathbb{E}[X]}{\Gamma} \right)^{\Gamma t} \leq \left(\frac{e^{(1-\frac{\mu}{\Gamma})} \mu}{\Gamma} \right)^{\Gamma t},$$

where the final inequality follows from the fact that x/e^x is decreasing in x . $\square$

We will also need the following analytic inequality.

Lemma 4.5. For every $c \geq 1$ and $x \geq 2$ we have

$$0 \leq e^{-1/c} - \left(1 - \frac{1}{cx}\right)^x \leq \frac{A}{x},$$

where $A = 5e^{-1/c}/c^2$.

Proof. Let $f(x) = (1 - 1/(cx))^x$. Then $\lim_{x \rightarrow \infty} f(x) = e^{-1/c}$. Additionally, we have the following technical claim, whose proof is deferred to [Appendix B](#) for clarity.

Claim 4.6.

$$f'(x) = f(x) \left(\ln\left(1 - \frac{1}{cx}\right) + \frac{1}{cx-1} \right) = f(x) \sum_{i=2}^{\infty} \frac{i-1}{i(cx)^i} > 0.$$

Therefore, $f(x)$ is a strictly monotone increasing function and $0 \leq e^{-1/c} - f(x)$. By the mean value theorem, there exists $\xi \in [1, 2]$ for which

$$f(2x) - f(x) = f'(\xi x)x = f(\xi x) \left(\ln\left(1 - \frac{1}{\xi cx}\right) + \frac{1}{\xi cx-1} \right)x.$$

We now use the inequalities $\ln(1-y) \leq -y + y^2/2$ and $1/(z-1) \leq 1/z + 2/z^2$ for any $y < 1$ and $z > 2$ and the fact that $f(x) = (1 - 1/(cx))^x$ is a strictly monotone increasing function and get

$$f(2x) - f(x) \leq f(\xi x) \left(-\frac{1}{\xi cx} + \frac{1}{2\xi^2 c^2 x^2} + \frac{1}{\xi cx} + \frac{2}{\xi^2 c^2 x^2} \right)x \leq \frac{2.5f(2x)}{c^2 x} \leq \frac{2.5e^{-1/c}}{c^2 x} = \frac{A}{2x}.$$

Therefore

$$e^{-1/c} - f(x) = \lim_{n \rightarrow \infty} f(2^n x) - f(x) = \sum_{i=1}^{\infty} f(2^i x) - f(2^{i-1} x) \leq \sum_{i=1}^{\infty} \frac{A}{2^i x} = \frac{A}{x}. \quad \square$$

4.2. Proof of the theorem

In this section, we prove [Theorem 4.1](#).

Proof. We will begin by presenting the proof for the case of $\lambda = 1$ and any α . This allows for a clearer understanding of the intuition behind the general proof, making it easier to follow the logic without getting lost in the details.

The proof concept in [\[6\]](#) for $\lambda = 1$ and any constant α relies on the following facts:

- (U1) $P(d, p)$ is a strictly monotone increasing function in p .
- (U2) For some constant $1 \leq c = \Theta_d(1)$ we have $\Delta(\alpha) := P(d, 1/(cd)) - P(d, \alpha^{-1/2}/(cd)) = \Theta_d(1)$. That is, $\Delta(\alpha)$ is greater than some constant that is independent of d (and n).
- (U3) For every constant $0 < \beta = \Theta_d(1)$ there is a constant $b = \Theta_d(1)$ such that for every $d \geq b$ we have $|P(d, 1/(cd)) - \lim_{x \rightarrow \infty} P(x, 1/(cx))| \leq \beta$.
- (U4) There is a constant $1 > c' = \Theta_d(1)$ such that for every $0 \leq x \leq 1/cd$ we have $P(d, x) \leq c'P(d, \alpha^{1/4}x)$.

The algorithm's core idea is to estimate $P(d, p)$ at specific geometric progression points $p = p_i := \alpha^{i/4}/U$ for $i = 0, 1, \dots, 4 \log_\alpha(U/L)$. We then select the first index i_1 such that $P(d, p_{i_1})$ closely approximates $P(d, 1/(cd))$ and employ p_{i_1} to estimate the value of d . Each estimation is constrained to use at most $O(\log(1/\delta))$ queries. To achieve this, we rely on the condition that $P(d, 1/(cd)) - P(d, 1/(\alpha^{1/2}cd))$ is constant (item (U2)). Since we lack knowledge of the true value of d , and therefore of $P(d, 1/(cd))$, we substitute it with $P(d', 1/(cd'))$, which is in proximity (see [\(8\)](#) that follows from item (U3)). The additional condition in item (U4) ensures that the estimations at $i = 0, 1, \dots, i_1$ can be performed with probability at least $1 - \delta$ and within the query limit of $O(\log(1/\delta))$ for each estimation.

Using item (U3) with $\beta = \min(\Delta(\alpha), 2)/16$, there is a constant d' such that for every $d \geq d'$ we have

$$|P(d, 1/(cd)) - \lim_{x \rightarrow \infty} P(x, 1/(cx))| \leq \Delta(\alpha)/16.$$

Therefore, for every $d \geq d'$ we have

$$|P(d', 1/(cd')) - P(d, 1/(cd))| \leq \Delta(\alpha)/8. \quad (8)$$

We first present the algorithm for estimating d for $\lambda = 1$, assuming d is greater than a sufficiently large constant d' . The algorithm estimates $P(d, \alpha^{i/4}/U)$ for all $i = 0, 1, \dots, 4 \log_\alpha(U/L)$, each with an additive error of at most $\Delta(\alpha)/8$ using $O(\log(1/\delta))$ queries (for each i).⁵ (Recall the algorithm is non-adaptive, so all queries must be chosen at once.) The algorithm selects the first index i_1 for which the estimated value of $P(d, \alpha^{i_1/4}/U)$ is greater than the benchmark value $P(d', 1/(cd')) - \Delta(\alpha)/4$ and returns $D = U/(c\alpha^{(i_1-1)/4})$ as the estimation.

Algorithm 1: Estimation of d

Data: Parameters $\alpha, L, U, \delta, c, d'$ and $\Delta(\alpha)$.

Result: Estimated value $d \leq D \leq \alpha d$.

- 1 For $i = 0, 1, \dots, 4 \log_\alpha(U/L)$;
 - 2 Estimate $P(d, \alpha^{i/4}/U)$ with an additive error of at most $\Delta(\alpha)/8$ using $O(\log(1/\delta))$ queries;
 - 3 Select the first i_1 such that the estimated value of $P(d, \alpha^{i_1/4}/U)$ is greater than $P(d', 1/(cd')) - \Delta(\alpha)/4$;
 - 4 **return** $D = U/(c\alpha^{(i_1-1)/4})$;
-

Now, assuming that items (U1)–(U4) hold, we proceed to establish the correctness of the algorithm. We will say that the estimation of $P(d, \alpha^{i/4}/U)$ is *correct* if it is within an additive error of at most $\Delta(\alpha)/8$ from $P(d, \alpha^{i/4}/U)$.

Claim 4.7. *If the first i_1 estimations of $P(d, \alpha^{i/4}/U)$ are correct, then $D \in [d, \alpha d]$.*

Proof. Suppose the first i_1 estimations are correct. Since the estimation of $P(d, \alpha^{i_1/4}/U)$ is greater than $P(d', 1/(cd')) - \Delta(\alpha)/4$, we have that $P(d, \alpha^{i_1/4}/U)$ is greater than $P(d', 1/(cd')) - 3\Delta(\alpha)/8$. Since by [\(8\)](#), $|P(d', 1/(cd')) - P(d, 1/(cd))| \leq \Delta(\alpha)/8$, we have that $P(d, \alpha^{i_1/4}/U) > P(d, 1/(cd)) - \Delta(\alpha)/2$. By item (U2),

$$P(d, \alpha^{i_1/4}/U) > P(d, 1/(cd)) - \Delta(\alpha)/2 = P(d, \alpha^{-1/2}/(cd)) + \Delta(\alpha)/2 > P(d, \alpha^{-1/2}/(cd))$$

and therefore by item (U1), $\alpha^{i_1/4}/U > \alpha^{-1/2}/(cd)$ and

$$D = \frac{U}{c\alpha^{(i_1-1)/4}} = \frac{\alpha^{1/4}}{c\alpha^{i_1/4}/U} < \frac{\alpha^{1/4}}{c(\alpha^{-1/2}/cd)} = \alpha^{3/4}d \leq \alpha d.$$

Let i_2 be an integer that satisfies $\alpha^{1/4}/(cd) \geq \alpha^{i_2/4}/U > 1/(cd)$. Then by [\(8\)](#) and item (U1), $P(d, \alpha^{i_2/4}/U) > P(d, 1/(cd)) \geq P(d', 1/(cd')) - \Delta(\alpha)/8$. The estimation of $P(d, \alpha^{i_2/4}/U)$ is greater than $P(d', 1/(cd')) - \Delta(\alpha)/4$. Therefore $i_1 \leq i_2$ and $\alpha^{i_1/4}/U \leq \alpha^{i_2/4}/U \leq \alpha^{1/4}/(cd)$. Thus $D = U/(c\alpha^{(i_1-1)/4}) \geq d$. $\square$

⁵ Claim [4.8](#) shows that this is possible for the first i_1 elements of $P(d, \alpha^{i/4}/L)$.

The estimation of $P(d, \alpha^{i/4}/U)$ can be accomplished using Chernoff bound, with each estimation requiring $O(\log(1/\delta))$ queries. We need item (U4) to show that, with probability at least $1 - \delta$, all the estimations up to i_1 are correct.

Claim 4.8. *The probability that all the estimations of $P(d, \alpha^{i/4}/U)$ for $i \leq i_1$ have an additive error of at most $\Delta(\alpha)/8$ is at least $1 - \delta$.*

Proof. Since by item (U4), $P(d, x) \leq c'P(d, \alpha^{1/4}x)$, $c' < 1$ and $P(d, \alpha^{i_1/4}/U) < 1$, there is a constant j_0 such that $P(d, \alpha^{(i_1-j_0)/4}/U) \leq \Delta(\alpha)/32$. Since j_0 and $\Delta(\alpha)/8$ are constants, we can estimate all $P(d, \alpha^{(i_1-j)/4}/U)$, $j = 0, 1, \dots, j_0$ with additive error of at most $\Delta(\alpha)/8$ and probability at least $1 - \delta/2$ with $O(\log(1/\delta))$ queries.

Since $P(d, \alpha^{(i_1-j-i)/4}/U) \leq \Delta(\alpha)/32$, by item (U4), we have $P(d, \alpha^{(i_1-j-i)/4}/U) \leq c'^i \Delta(\alpha)/32$, $i = 1, 2, \dots, i_1 - j$. Applying Lemma 4.4 with $\mu = P(d, \alpha^{(i_1-j-i)/4}/U)$, $\Gamma = \Delta(\alpha)/8$, and $t = O(\log(1/\delta))$ with some sufficiently large implicit constant, we find the probability that the estimation of $P(d, \alpha^{(i_1-j-i)/4}/U)$ has additive error greater than $\Delta(\alpha)/8$ is at most

$$\left(\frac{eP(d, \alpha^{(i_1-j-i)/4}/U)}{\Delta(\alpha)/8} \right)^{\frac{\Delta(\alpha)}{8} O(\log(\frac{1}{\delta}))} \leq \left(\frac{e(c'^i \Delta(\alpha)/32)}{\Delta(\alpha)/8} \right)^{O(\log(\frac{1}{\delta}))} \leq \left(\frac{ec'^i}{4} \right)^{O(\log(\frac{1}{\delta}))} \leq \left(\frac{\delta}{4} \right)^{i+1}.$$

The probability that one of the estimations of $P(d, 1/U)$, $P(d, \alpha^{1/4}/U)$, $\dots$, $P(d, \alpha^{(i_1-j_0)/4}/U)$ has additive error greater than $\Delta(\alpha)/8$ is at most

$$\sum_{i=0}^{i_1-j_0} \left(\frac{\delta}{4} \right)^{i+1} \leq \frac{\delta}{2}. \quad \square$$

We now prove items (U1)–(U4). Item (U1) is clear from its formula. Since P is a strictly monotone increasing function, if items (U2)–(U4) hold for $\alpha = \alpha'$, they also hold for any $\alpha \geq \alpha'$. Therefore, it is sufficient to prove that items (U2)–(U4) hold for any constant $1 < \alpha \leq 2$.

Since $P(d, p) = 1 - (1 - p)^d$, by the mean value theorem, Lemma 4.5 and since $\frac{\partial}{\partial p} P(d, p) = d(1 - p)^{d-1}$, there is $1/(cd) \geq \eta \geq \alpha^{-1/2}/(cd)$ such that

$$\begin{aligned} \Delta(\alpha) &= P(d, 1/(cd)) - P(d, \alpha^{-1/2}/(cd)) &< \text{By definition} \\ &= d(1 - \eta)^{d-1}(1/(cd) - \alpha^{-1/2}/(cd)) &< \text{By mean value theorem} \\ &\geq \left(1 - \frac{1}{cd}\right)^{d-1} (1 - \alpha^{-1/2})/c &< \text{Since } \eta \leq 1/(cd) \\ &\geq \left(1 - \frac{1}{c(d-1)}\right)^{d-1} (1 - \alpha^{-1/2})/c \\ &\geq (e^{-1/c} - A/(d-1))(1 - \alpha^{-1/2})/c &< \text{Lemma 4.5} \\ &= \Theta_d(1). &< d \geq d' \text{ for large enough constant } d'. \end{aligned}$$

This implies item (U2) for any constant c .

We now prove item (U3). Given a constant β , let $b = \lceil A/\beta \rceil$ where A is the constant in Lemma 4.5. By Lemma 4.5, for $d \geq b$, we have

$$0 \leq P(d, 1/(cd)) - (1 - e^{-1/c}) \leq \frac{A}{d} \leq \frac{A}{b} \leq \beta.$$

Since

$$\lim_{x \rightarrow \infty} P(x, 1/cx) = \lim_{x \rightarrow \infty} 1 - \left(1 - \frac{1}{cx}\right)^x = 1 - e^{-1/c}$$

the result follows.

We will now prove item (U4). Let $c = \alpha$. Since for $0 \leq x \leq 1$ we have $1 - dx \leq (1 - x)^d \leq 1 - dx + d^2x^2/2$, it follows that $P(d, x) = 1 - (1 - x)^d \leq dx$. Since $0 \leq x \leq 1/cd$ and $c = \alpha$, we have

$$P(d, \alpha^{1/4}x) = 1 - (1 - \alpha^{1/4}x)^d \geq \alpha^{1/4}dx - d^2\alpha^{1/2}x^2/2 \geq (\alpha^{1/4} - \alpha^{-1/2})dx \geq (\alpha^{1/4} - \alpha^{-1/2})P(d, x).$$

This completes the proof for $\lambda = 1$ and any $\alpha = 1 + \Omega(1)$. Now, we will extend the proof to cover any constant $\lambda > 1$ and $\alpha = 1 + \Omega(1)$.

It is easy to verify that

$$\frac{\partial}{\partial p} P_\lambda(d, p) = \left(\frac{d}{\lambda - 1} \right) (d - \lambda + 1) p^{\lambda-1} (1 - p)^{d-\lambda}.$$

To get the result, we show

(V1) $P_\lambda(d, p)$ is a strictly monotone increasing function in p .

(V2) For some constant $c \geq 1$, we have $\Delta_\lambda(\alpha) := P_\lambda(d, \lambda/(cd)) - P_\lambda(d, \alpha^{-1/2}\lambda/(cd)) = \Theta_d(1)$.

(V3) For every constant β , there is a constant d' such that for every $d \geq d'$ we have

$$\left| P_\lambda(d, \lambda/(cd)) - \lim_{x \rightarrow \infty} P_\lambda(x, \lambda/(cx)) \right| \leq \beta.$$

(V4) There is a constant $c' < 1$ such that for every $0 \leq x \leq \lambda/cd$ we have $P_\lambda(d, x) \leq c'P_\lambda(d, \alpha^{1/4}x)$.

The algorithm and its correctness for general λ follow similarly as in the case of $\lambda = 1$, with P replaced by P_λ and c replaced by c/λ . It remains to prove items (V1)–(V4) for $\lambda > 1$.

Item (V1) follows because $\frac{\partial}{\partial p} P_\lambda(d, p) > 0$ for all $0 < p \leq 1$. Since P_λ is a strictly monotone increasing function, if items (V2)–(V4) hold for $\alpha = \alpha'$, then they also hold for any $\alpha \geq \alpha'$. Therefore, it is enough to prove items (V2) and (V4) for any constant $1 < \alpha \leq 2$.

Let $c = \frac{2\lambda}{1-\alpha^{-1/4}}$. By mean value theorem, there is $1/\alpha^{1/2} \leq \eta \leq 1$ such that

$$\begin{aligned} \Delta_\lambda(\alpha) &= P_\lambda(d, \lambda/(cd)) - P_\lambda(d, \lambda/(\alpha^{1/2}cd)) \\ &= \binom{d}{\lambda-1} (d-\lambda+1) \left(\frac{\eta\lambda}{cd} \right)^\lambda \left(1 - \frac{\eta\lambda}{cd} \right)^{d-\lambda} \left(\frac{\lambda}{cd} - \frac{\lambda}{\alpha^{1/2}cd} \right) \\ &= \frac{d(d-1)\cdots(d-\lambda+1)}{d^{\lambda+1}} \frac{\lambda^{\lambda+1}}{(\lambda-1)!} \frac{\eta^\lambda}{c^{\lambda+1}} \left(1 - \frac{\eta\lambda}{cd} \right)^{d-\lambda} (1 - \alpha^{-1/2}) \\ &\geq 2^{\lambda+1} \frac{\lambda^{\lambda+1}}{(\lambda-1)!} \frac{\eta^\lambda}{c^{\lambda+1}} \frac{1}{2} (1 - \alpha^{-1/2}) < \lambda < d/2, c > 2\lambda \\ &= \Theta_d(1). \end{aligned}$$

This proves item (V2).

We now prove item (V3). First, since λ is constant,

$$\lim_{x \rightarrow \infty} P_\lambda(x, \lambda/(cx)) = 1 - \sum_{i=0}^{\lambda-1} \left(\frac{\lambda}{c} \right)^i \frac{e^{-\lambda/c}}{i!}.$$

For the proof we will use the following inequalities: For any $i < \lambda$ we have

$$1 \geq \frac{d(d-1)(d-2)\cdots(d-i+1)}{d^i} \geq 1 - \frac{\lambda^2}{2d} \quad \text{and} \quad 1 \geq \left(1 - \frac{\lambda}{cd} \right)^i \geq 1 - \frac{\lambda^2}{cd}.$$

Now, by Lemma 4.5 and the above inequalities,

$$\begin{aligned} \left| P_\lambda(d, \lambda/(cd)) - \left(1 - \sum_{i=0}^{\lambda-1} \left(\frac{\lambda}{c} \right)^i \frac{e^{-\lambda/c}}{i!} \right) \right| &= \left| \sum_{i=0}^{\lambda-1} \left(\left(\frac{\lambda}{c} \right)^i \frac{e^{-\lambda/c}}{i!} - \binom{d}{i} \left(\frac{\lambda}{cd} \right)^i \left(1 - \frac{\lambda}{cd} \right)^{d-i} \right) \right| \\ &= \left| \sum_{i=0}^{\lambda-1} \left(\left(\frac{\lambda}{c} \right)^i \frac{1}{i!} \left(e^{-\lambda/c} - \frac{\prod_{\ell=0}^{i-1} (d-\ell)}{d^i} \frac{(1-\frac{\lambda}{cd})^d}{(1-\lambda/cd)^i} \right) \right) \right| \\ &\leq \sum_{i=1}^{\lambda-1} \left(\frac{\lambda}{c} \right)^i \frac{1}{i!} \left| e^{-\lambda/c} - \left(1 \pm O\left(\frac{1}{d} \right) \right) \left(1 - \frac{\lambda}{cd} \right)^d \right| \\ &\leq \sum_{i=1}^{\lambda-1} \left(\frac{\lambda}{c} \right)^i \frac{1}{i!} \left| e^{-\lambda/c} - \left(1 - \frac{\lambda}{cd} \right)^d \pm O\left(\frac{1}{d} \right) \right| \\ &= O\left(\frac{1}{d} \right). \end{aligned}$$

This proves item (V3).

We now prove item (V4). For every $0 \leq j \leq d-\lambda$ and $0 \leq x \leq \lambda/cd$, we have

$$\begin{aligned} \frac{\binom{d}{\lambda+j} x^{\lambda+j} (1-x)^{d-\lambda-j}}{\binom{d}{\lambda+j} (\alpha^{1/4}x)^{\lambda+j} (1-\alpha^{1/4}x)^{d-\lambda-j}} &\leq \frac{1}{\alpha^{\lambda/4} (1-\alpha^{1/4}x)^d} \\ &\leq \frac{1}{\alpha^{\lambda/4} (1-\alpha^{1/4}xd)} \end{aligned}$$

$$\begin{aligned}
&\leq \frac{1}{\alpha^{\lambda/4}(1 - \alpha^{1/4}\lambda/c)} \\
&\leq \frac{1}{\alpha^{(\lambda-1)/4}} \cdot \quad \triangleleft c > \frac{\alpha^{1/4}\lambda}{1 - \alpha^{-1/4}}
\end{aligned}$$

Therefore, by (7),

$$P_\lambda(d, x) = \sum_{i=\lambda}^d \binom{d}{i} x^i (1-x)^{d-i} \leq \frac{1}{\alpha^{(\lambda-1)/4}} \sum_{i=\lambda}^d \binom{d}{i} (\alpha^{1/4}x)^i (1 - \alpha^{1/4}x)^{d-i} \leq \frac{1}{\alpha^{(\lambda-1)/4}} P_\lambda(d, \alpha^{1/4}x).$$

This implies item (V4) and the result follows. $\square$

5. Concluding remarks

This work illustrates the power of the distribution distinguishing problem as a lower-bound technique for the group testing estimation problem. The main result uses a fairly straightforward coupling, which simply matches the planted defect set sizes within a β -factor, and this coupling already yields the desired tight lower bound for our case. Therefore it is reasonable to assert that the potential of this technique is not fully explored. One possible future direction is to extend this technique to prove lower bounds for other query models such as gap-threshold queries [17] and density tests [1].

Acknowledgements

This work was done in part during TMC, HH, and AO's visit at the Simons Institute for the Theory of Computing. We thank anonymous reviewers for useful feedback on an earlier version of this paper.

Appendix A. Exploring coupling: basic concepts

In this appendix, we provide basic definitions and results for readers who may not be familiar with the coupling technique. To ensure completeness, we begin with fundamental concepts in probability.

Let S be a set. We say that $\mathcal{F}$ is σ -algebra on S if $\mathcal{F} \subseteq 2^S$ is a set of subsets of S and $\mathcal{F}$ is closed under complement, unions and countable intersections. The pair $(S, \mathcal{F})$ is called a *measurable space*. A *measure* on a measurable space $(S, \mathcal{F})$ is a function $\mu : \mathcal{F} \rightarrow \mathbb{R} \cup \{-\infty, +\infty\}$ such that: (1) for every $F \in \mathcal{F}$, $\mu(F) \geq 0$. (2) $\mu(\emptyset) = 0$ and (3) for any $\{F_i\}_{i \in \mathbb{N}}$, $F_i \in \mathcal{F}$, $\mu(\bigcup_{i=1}^{\infty} F_i) = \sum_{i=1}^{\infty} \mu(F_i)$. The tuple $(S, \mathcal{F}, \mu)$ is called a *measure space*. If $\mu(S) = 1$ then μ is called a *probability measure* and $(S, \mathcal{F}, \mu)$ is called a *probability space*.

Given two measurable spaces $(S_1, \mathcal{F}_1)$ and $(S_2, \mathcal{F}_2)$. A function $X : S_1 \rightarrow S_2$ is a *measurable function* if for every $F \in \mathcal{F}_2$, $X^{-1}(F) := \{\omega \in S_1 | X(\omega) \in F\} \in \mathcal{F}_1$. When $(S_1, \mathcal{F}_1, \mu)$ is a probability space, then X is called a *random variable*. For such X , we say that X is *defined on* $(S_1, \mathcal{F}_1)$ and *takes values in* $(S_2, \mathcal{F}_2)$. The *law of* X , denoted by μ_X , is a probability measure on $(S_2, \mathcal{F}_2)$ defined as: For every $F \in \mathcal{F}_2$ $\mu_X(F) = \mu(X^{-1}(F))$.

Now, we give the formal definition of coupling.

Definition A.1 (Coupling). Let μ_1 and μ_2 be probability measures on the same measurable space $(S, \mathcal{F})$. A *coupling* of μ_1 and μ_2 is a probability measure μ on the product space⁶ $(S \times S, \mathcal{F} \times \mathcal{F})$ such that for every $F \in \mathcal{F}$, we have $\mu(F \times S) = \mu_1(F)$ and $\mu(S \times F) = \mu_2(F)$.

For two random variables X and Y taking values in $(S, \mathcal{F})$ (but not necessarily defined on the same probability space), a coupling of X and Y is a joint variable (X', Y') taking values in $(S \times S, \mathcal{F} \times \mathcal{F})$ where⁷ $\mu_{(X', Y')}$ is a coupling of μ_X and μ_Y . We also say that (X', Y') is a coupling of μ_1 and μ_2 if $\mu_{(X', Y')}$ is a coupling of μ_1 and μ_2 .

A.1. Basic results

In this section, we present two well-known results that will be used in the paper, along with their proofs for completeness.

Lemma A.2. Let $(S, \mathcal{F}, \mu)$ be a probability space, $(S_1, \mathcal{F}_1)$ and $(S_2, \mathcal{F}_2)$ measurable spaces, $X', Y' : S \rightarrow S_1$ random variables, X and Y are random variables that take values in $(S_1, \mathcal{F}_1)$ and $f : S_1 \rightarrow S_2$ a measurable function such that $f(S_1) = S_2$. If (X', Y') is a coupling of X and Y then $(f(X'), f(Y'))$ is a coupling of $f(X)$ and $f(Y)$.

⁶ Here $\mathcal{F} \times \mathcal{F}$ is the smallest σ -algebra that contains the Cartesian product of $\mathcal{F}$ with itself.

⁷ $\mu_{(X', Y')}$ is the law of (X', Y') .

Proof. It is clear that $f(X'), f(Y') : S \rightarrow S_2$ are random variables that take values in $(S_2, \mathcal{F}_2)$ and $f(X)$ and $f(Y)$ take values in $(S_2, \mathcal{F}_2)$. Now for any $F \in \mathcal{F}_2$,

$$\begin{aligned}\mu_{(f(X'), f(Y'))}(F \times S_2) &= \mu(\{\omega \in S \mid f(X'(\omega)) \in F, f(Y'(\omega)) \in S_2\}) \\ &= \mu(\{\omega \in S \mid X'(\omega) \in f^{-1}(F), Y'(\omega) \in f^{-1}(S_2)\}) \\ &= \mu_{(X', Y')}(f^{-1}(F) \times f^{-1}(S_2)) \\ &= \mu_{(X', Y')}(f^{-1}(F) \times S_1) = \mu_X(f^{-1}(F)) = \mu_{f(X)}(F).\end{aligned}$$

In the same way $\mu_{(f(X'), f(Y'))}(S_2 \times F) = \mu_{f(Y)}(F)$. $\square$

Lemma A.3. Let μ_1 and μ_2 be two probability measures on the measurable space $(S, \mathcal{F})$. For any coupling (X, Y) of μ_1 and μ_2 we have

$$\|\mu_1 - \mu_2\|_{TV} \leq \Pr[X \neq Y]$$

where⁸ $[X \neq Y] = \{\omega \in S \mid X(\omega) \neq Y(\omega)\}$ and X and Y are random variables defined on the probability space $(S', \mathcal{F}', \Pr)$ (and take values in $(S, \mathcal{F})$).

Proof. Let $[X \neq Y] = \{\omega \in S \mid X(\omega) \neq Y(\omega)\}$. Since for any $B \in \mathcal{F}$ we have $X^{-1}(B) \cap [X = Y] = Y^{-1}(B) \cap [X = Y]$, for any $A \in \mathcal{F}$, we have

$$\begin{aligned}\mu_1(A) - \mu_2(A) &= \mu_{X \times Y}(A \times S) - \mu_{X \times Y}(S \times A) \\ &= \Pr[\{\omega \in S' \mid X(\omega) \in A, Y(\omega) \in S\}] - \Pr[\{\omega \in S' \mid X(\omega) \in S, Y(\omega) \in A\}] \\ &= \Pr[X^{-1}(A)] - \Pr[Y^{-1}(A)] \\ &= \Pr[X^{-1}(A) \cap [X = Y]] + \Pr[X^{-1}(A) \cap [X \neq Y]] \\ &\quad - \Pr[Y^{-1}(A) \cap [X = Y]] - \Pr[Y^{-1}(A) \cap [X \neq Y]] \\ &= \Pr[X^{-1}(A) \cap [X \neq Y]] - \Pr[Y^{-1}(A) \cap [X \neq Y]] \\ &\leq \Pr[X \neq Y]. \quad \square\end{aligned}$$

While unnecessary for understanding this work, the interested reader may wish to consult [19] or [29, Chapter 4] for a more extensive treatment of coupling.

Appendix B. Proof of Claim 4.6

Given the function:

$$f(x) = \left(1 - \frac{1}{cx}\right)^x,$$

we want to find its derivative, $f'(x)$. First, taking the natural logarithm of $f(x)$ gives

$$\ln f(x) = x \ln \left(1 - \frac{1}{cx}\right).$$

Using the chain rule to differentiate with respect to x :

$$\frac{f'(x)}{f(x)} = \ln \left(1 - \frac{1}{cx}\right) + x \cdot \frac{d}{dx} \left[\ln \left(1 - \frac{1}{cx}\right) \right].$$

The derivative of $\ln \left(1 - \frac{1}{cx}\right)$ is

$$\frac{d}{dx} \left[\ln \left(1 - \frac{1}{cx}\right) \right] = \frac{1}{1 - \frac{1}{cx}} \cdot \left(\frac{1}{cx^2} \right).$$

Therefore

$$\frac{f'(x)}{f(x)} = \ln \left(1 - \frac{1}{cx}\right) + \frac{1}{cx - 1}$$

and

$$f'(x) = f(x) \left(\ln \left(1 - \frac{1}{cx}\right) + \frac{1}{cx - 1} \right).$$

⁸ Here we also assume that $[X \neq Y] \in \mathcal{F}'$.

Now, using the Taylor expansion of $\ln(1 - u)$ around $u = 0$:

$$\ln\left(1 - \frac{1}{cx}\right) = -\frac{1}{cx} - \frac{1}{2(cx)^2} - \frac{1}{3(cx)^3} - \dots$$

and

$$\frac{1}{cx - 1} = \frac{1}{cx} + \frac{1}{(cx)^2} + \dots$$

we get

$$f'(x) = f(x) \sum_{i=2}^{\infty} \frac{i-1}{i(cx)^i}.$$

Since each term $\frac{i-1}{i(cx)^i}$ is positive for all $i \geq 2$ and $cx > 0$, the entire sum is positive. Therefore,

$$f'(x) > 0.$$

This shows that $f(x)$ is increasing for all sufficiently large x .

Data availability

No data was used for the research described in the article.

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